

How to Find an Example

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Outline

Magic Squares

Narrowing the Search

Multiplicative Magic Squares

More Puzzles

Integer linear combinations

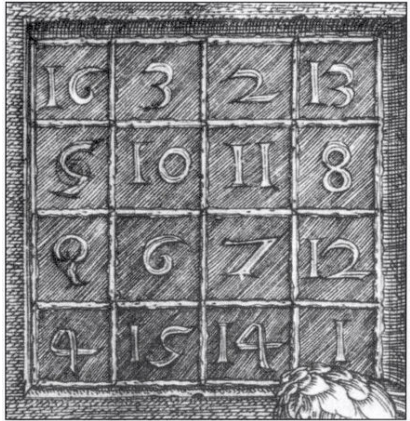
Paths in a Graph

Be creative!

Albrecht Duerer,
Melancholia, 1514

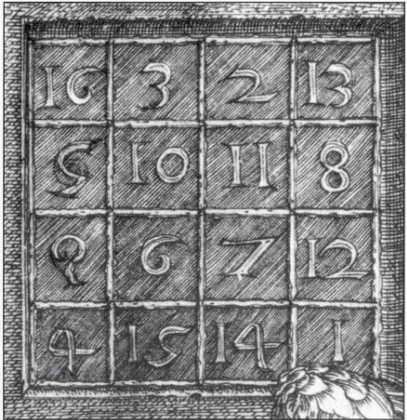


Magic Square: Definition



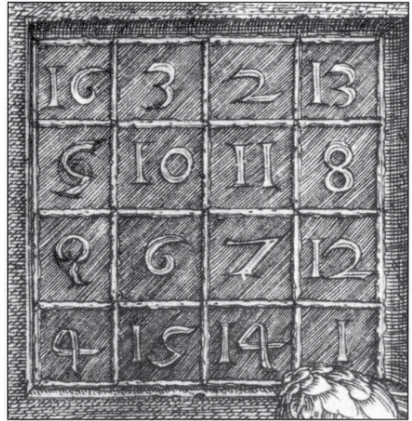
Magic Square: Definition

- 1, 2, 3, ..., 15, 16



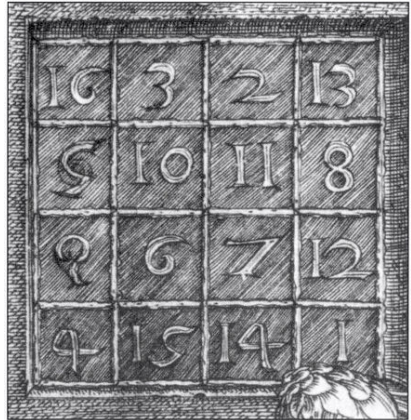
Magic Square: Definition

- $1, 2, 3, \dots, 15, 16$
- $1, 2, 3, \dots, n^2$
for $n \times n$



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- $1, 2, 3, \dots, 15, 16$
- $1, 2, 3, \dots, n^2$
for $n \times n$
- the same sum in
columns, rows,
diagonals



How to Find Magic Squares

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– why?

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- what about size 3 – made of $1, 2, 3, \dots, 9$?

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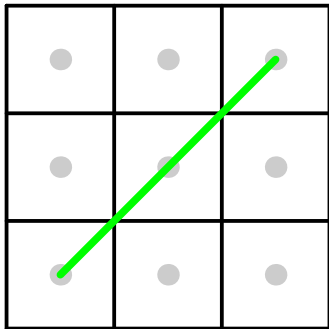
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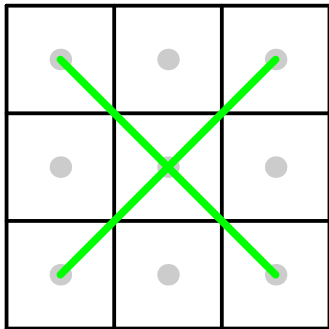
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- $45/3 = 15$

Hint: the Center



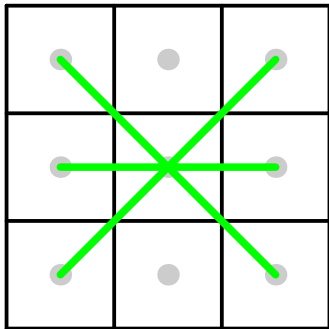
summing up four lines...

Hint: the Center



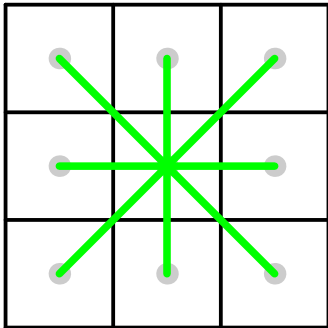
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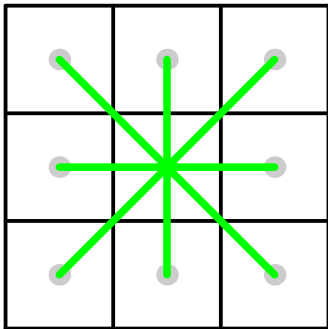
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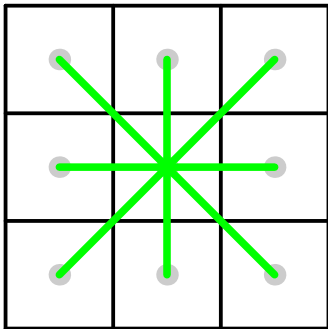
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$$4S = \text{total sum} + 3 \cdot \text{center}$$

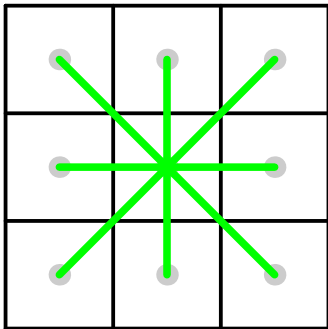
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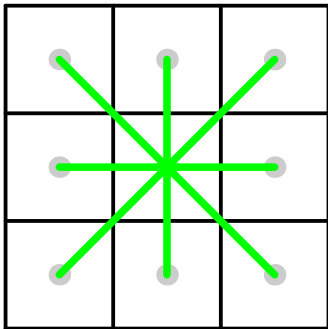
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summing up four lines...

$$S/3 = \text{center}$$

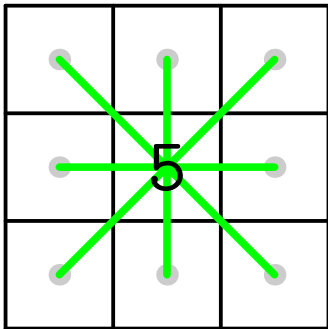
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summing up four lines...

$$S/3 = \text{center} = 5$$

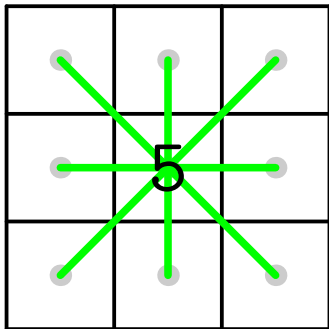
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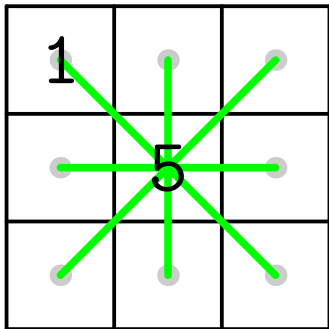
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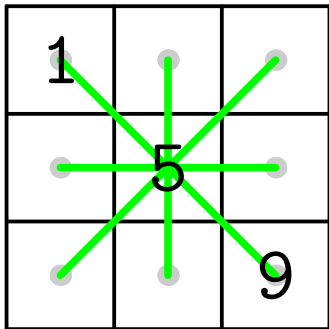
1: Corner or Middle?



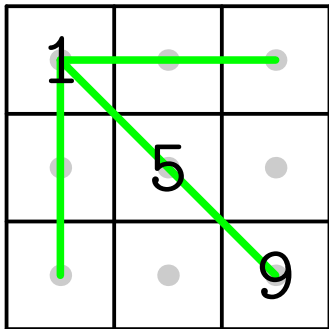
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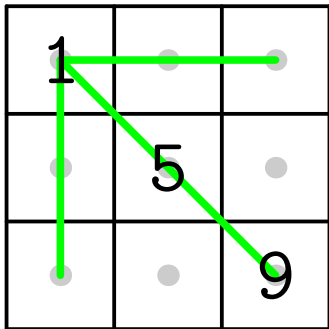
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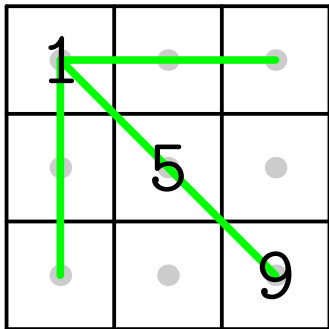


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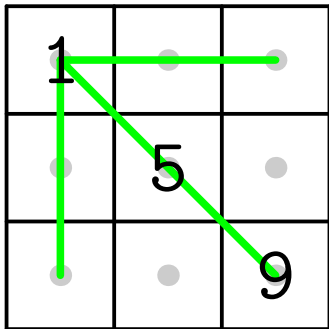
$$14 = 5 + 9$$

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$$14 = 5 + 9 = 6 + 8$$

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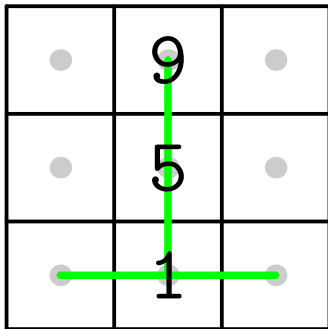


$$14 = 5 + 9 = 6 + 8 = 7 + 7$$

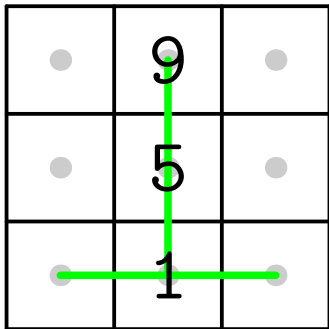
1: Middle!

•	9	•
•	5	•
•	1	•

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$$14 = 5 + 9 = 8 + 6$$

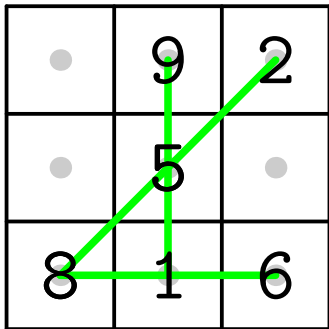
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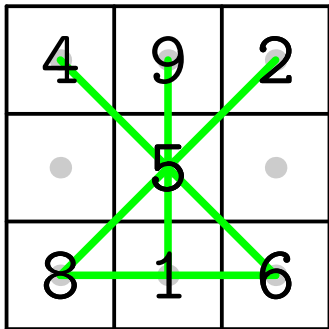
1: Middle!

•	9	2
•	5	•
8	1	6



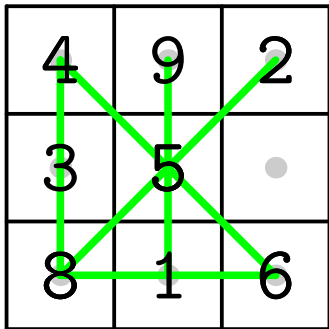
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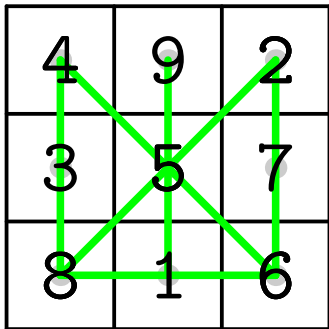
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- 7 appears in some products but not in the others
- arbitrary different positive integers allowed
- is it possible?

Spoiler

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3	5	7
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$\rightarrow$

2^4	2^9	2^2
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- numbers rather big — find smaller ones?
- less than 300?
- divide by 2
- less than 40?

Spoiler-2

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1	2	0
0	1	2
2	0	1

→

2	4	1
1	2	4
4	1	2

Spoiler-2

1	2	0
0	1	2
2	0	1

→

2	4	1
1	2	4
4	1	2

0	2	1
2	1	0
1	0	2

→

1	9	3
9	3	1
3	1	9

Spoiler-2

1	2	0
0	1	2
2	0	1

→

2	4	1
1	2	4
4	1	2

×

0	2	1
2	1	0
1	0	2

→

1	9	3
9	3	1
3	1	9

2	36	3
9	6	4
12	1	18

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- $11 \times 9127 = 100\,397$

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- next one is too big

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- $2 \times 2 \times 3 \times 5 \times 7 = 420$

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- $N - 1 = 420$

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- $N - 1 = 420$; $N = 421$

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- others? $420 \cdot 2 + 1 = 841$

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- 420×3 is too big

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- $177\,244^2 = 31\,415\,435\,536$

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- $560491^2 = 314\,150\,161\,081$: OK

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- can we use the same trick?
- $\sqrt{3141.5} = 56.0490856 \dots$
- $5605^2 = 31\,416\,025$: too big
- $560491^2 = 314\,150\,161\,081$: OK
- answer: 177 243 and 560 491.

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7 and 13

Imagine a country with $7/13$ florins coins.
Two people, each has many coins of each type.
Can one pay 6 florins to the other?

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- $6 = 13 - 7$
- what about paying 1 florin?

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- for every integer c , the equation $7x + 13y = c$ has integer solutions

15 **and** 21

15/21 coins; how to pay 6?

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- 3? OK: $6 = 21 - 15$; $9 = 15 - 6$, $3 = 9 - 6$;

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 $3 = 3 \times 15 - 2 \times 21$
- any multiple of 3 is payable
- the equation $15x + 21y = c$ has integer solutions $\Leftrightarrow c$ is a multiple of 3

Outline

Magic Squares

Narrowing the Search

Multiplicative Magic Squares

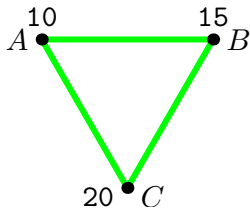
More Puzzles

Integer linear combinations

Paths in a Graph

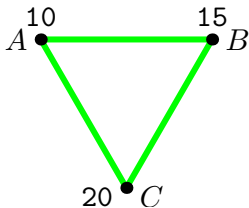
Hotels and Paths

- Hotels $A(10)$, $B(15)$, $C(20)$; change every night for $10 + 15 + 20 = 45$ nights



Hotels and Paths

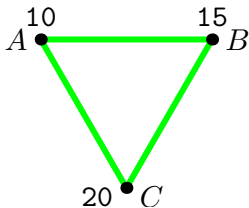
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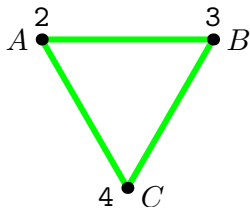
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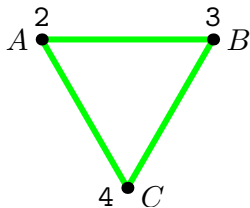
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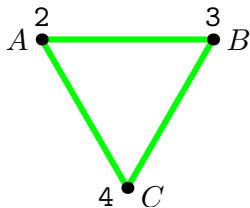
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- every second point is C

Hotels and Paths

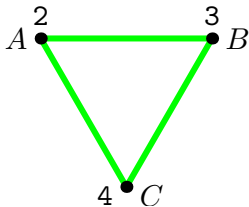
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- every second point is C
- A

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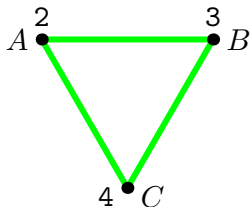
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- repeat 5 times a path of length 9
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- AC

Hotels and Paths

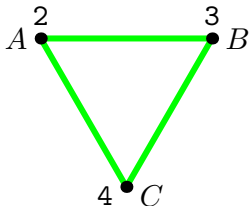
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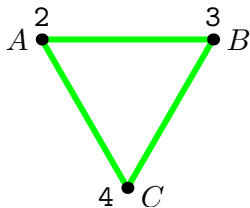
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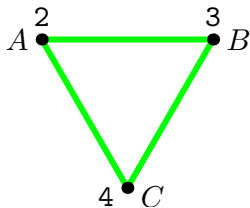
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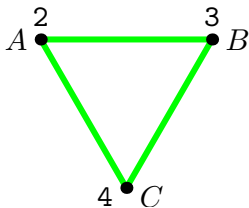
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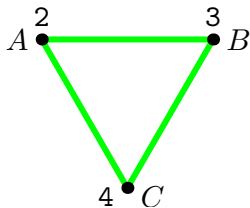
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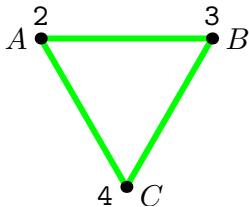
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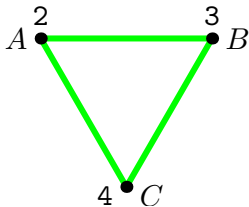
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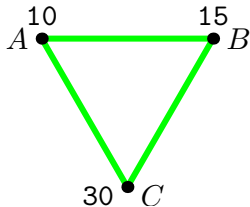
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- $ACACBCBCB$ (5 times)

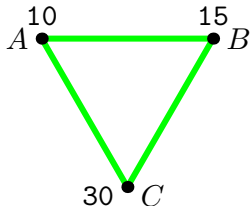
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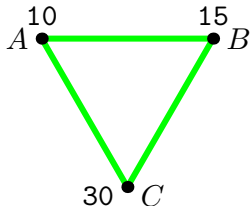
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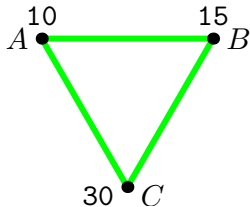
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- to use them all we need at least 29 others